STEAM SCOUTS

THE VISION WIZARDS

Grade 05

ALIGNED WITH INTERNATIONAL STANDARDS

This book supports essential educational frameworks that shape future-ready minds:

- 21st Century Skills
- Sustainable Development Goals (SDGs)
- Social-Emotional Learning (SEL)
- Universal Design for Learning (UDL)

STEAM Scouts is a ground-breaking STEAM book series with an interdisciplinary focus designed to ignite curiosity, creativity, and critical thinking in learners worldwide. The series supports Social-Emotional Learning (SEL) by promoting core competencies like self-awareness, self-management, social awareness, relationship skills, and responsible decision-making. It also aligns with the UN Sustainable Development Goals (SDGs), fosters 21st Century Skills, and meets international educational standards to equip learners with tools for a dynamic, innovative, and impact-driven future. Furthermore, it is built on Universal Design for Learning (UDL) principles, ensuring inclusive, accessible, and engaging STEAM education for all students, regardless of learning ability.

Key Features

Standards-Aligned

- NGSS (Next Generation Science Standards) – Authentic scientific inquiry & engineering design.
- CCSS (Common Core State Standards) – Rigorous literacy & math integration.
- Cambridge & IB (PYP/MYP/DP) – Global perspectives & interdisciplinary learning.
- ISTE (Technology Standards) – Digital literacy & computational thinking.
- STEL (Standards for Technological & Engineering Literacy) – Real-world problem-solving.

Social-Emotional Learning (SEL) and Global Citizenship

- Each book fosters collaboration, resilience, leadership, and ethical decision-making while connecting to multiple UN Sustainable Development Goals (SDGs) - such as Quality Education (SDG 4), Clean Energy (SDG 7), Climate Action (SDG 13), and Innovation (SDG 9) - empowering students to think globally, encourages ethical innovation & community impact.
- Differentiated challenges to meet diverse learning needs and skill levels.
- Visual cues and infographics to enhance understanding and engagement.
- Guided instruction to support learners through new or complex concepts.
- Scaffolded activities that build skills progressively and promote learner independence.

Why Choose This Series?

Future-Ready Skills – Prepares learners with essential skills like creativity, collaboration, and critical thinking for future careers and global challenges.

Teacher & Student-Friendly – Includes engaging lesson hooks, discussion prompts, guided instruction, and hands-on tasks for seamless learning.

- SDG-Themed Modules – Each module links to 2+ UN SDGs, blending academics with global purpose and ethical action.
- Inclusive & Engaging – Built on UDL principles to ensure accessibility through differentiated pathways, visual supports, and hands-on learning.

Perfect For:

- K-12 Schools adopting IB, Cambridge, NGSS, or CCSS
- STEM/STEAM Coordinators seeking industry-relevant curricula
- Homeschoolers and Educators wanting SDG-aligned, future-focused content
- Educational Institutions committed to sustainability and inclusion

Theme

03

THE VISION WIZARDS

Each week has a double session, total time of session 80-90 mins.

Week	Unit Name	Lesson	Page
Week 1	Unit 1: Spark The Challenge!	Lesson 1: The Incident at the Fair.	2-21
Week 2		Lesson 2: Discovering Wonders of the Eye.	22-39
Week 3		Lesson 3: Exploring Smart Glasses.	40-49
Week 4		Lesson 4: Unlocking Ultrasonic Secrets.	50-72
Week 5	Unit 2: Innovators in Action.	Lesson 1: Plan Your Perfect Screen!	73-91
Week 6		Lesson 2: Talk to Me!	92-108
Week 7	Unit 3: Beyond Imagination.	Lesson 1: Tinker Time.	109-122
Week 8		Lesson 2: Boost Your App's Senses.	123-140

About The Theme

Every person deserves the freedom to move safely, explore their surroundings, and live with confidence—regardless of their abilities. Yet for individuals with visual impairments, navigating busy or unfamiliar environments can be challenging. In this theme, students will become engineers, designers, and changemakers as they create smart assistive glasses that help people with vision loss "see" using sound, sensors, and intelligent technology.

Students will explore how science, coding, and artificial intelligence can enhance accessibility and independence. They'll learn how the human eye works, what causes vision loss, and how ultrasonic waves can detect objects—just like echolocation in bats and dolphins. Using ultrasonic sensors, Arduino, Bluetooth modules, and a custom mobile app, students will build a prototype that transforms their ideas into a functioning assistive device.

As they develop the mobile app with MIT App Inventor, students will use text-to-speech technology, a real-world application of AI, to provide voice-based instructions and feedback—helping users safely navigate their environment.Throughout the project, students will apply the Engineering Design Process to design, test, and refine their glasses. They'll model physical components in Tinkercad, program interactive behavior using block coding, and bring together hardware and software in a cohesive solution.

This project aligns with Sustainable Development Goals 3 (Good Health and Well-being) and 10 (Reduced Inequalities), encouraging students to think with empathy, innovate with purpose, and use technology for social good. By the end of the theme, students will have built a working, voice-guided wearable—and discovered how AI, coding, and design can empower everyone to experience the world more freely and confidently.

HOW TO USE THIS BOOK?

1

Ready, Set, Learn!

Where we merge your existing knowledge with exciting new discoveries.

2

Inquiry-Based Learning!

Let's Think!

In this section, students will brainstorm and explore possible solutions to specific problems. This encourages critical thinking and creativity as they consider various approaches.

Explorer's Toolkit

Here, students will dive deeper into the topic. They will engage with curated videos, articles, educational games, and hands-on experiments to broaden their understanding and spark curiosity.

DIY Lab

It's part of the explore, students will undertake hands-on projects where they will design, build, and test their own creations. They will apply scientific principles, use technology and engineering skills, explore artistic concepts, and utilize mathematical reasoning.

Screen Lessons

Students will watch videos that explain key concepts in an engaging and accessible way.

Dig Deeper

Students will read articles that provide deeper insights into specific concepts, enhancing their understanding.

3

Problem-Based Learning:

- Introducing the problem:
  Students will join a journey to tackle real-world environmental challenges. Through hands-on activities and engaging tasks, they will apply their knowledge and skills to discover innovative solutions.
- Apply the Engineering Design Process (EDP):
  Students will use a structured approach to define problems, conduct research, generate ideas, prototype solutions, test, evaluate, and refine their designs.
- The Scientific Method:
  Students will follow a systematic process of inquiry—making observations, forming hypotheses, conducting experiments, analyzing data, and drawing conclusions—to develop a deeper understanding of scientific concepts and problem-solving. In doing so, they will incorporate the CER framework to present their scientific claims, provide evidence from their experiments, and explain their reasoning behind the conclusions they draw.

4

Project-Based Learning:

PBL immerses students in real-world challenges through hands-on projects. They build critical thinking, creativity, collaboration, and communication skills while designing solutions and taking ownership of their learning.

5

Story

Join Adam and Laila, two fearless STEAM Scouts, on adventures full of discovery and creativity! From building inventions to solving real-world challenges, they'll inspire you to think, experiment, and innovate. Step into the world of science, technology, engineering, arts, and math—where every challenge becomes an adventure!

Story Reflection

In this section, students will reflect on the story's key themes, characters, and challenges.

6

Assessment Exercises:

These are divided into three categories to ensure comprehensive learning and mastery.

Apprentice:

Developing understanding and can apply skills with guidance.

Builder:

Demonstrates solid understanding and can apply skills independently.

Mastermind:

Demonstrates deep understanding and can apply skills creatively and strategically.

7

Career Readiness

This section equips students with essential skills for future careers by fostering problem-solving, teamwork, and adaptability. Through hands-on projects, real-world applications, and critical thinking challenges, students develop the confidence and competencies needed for success in the ever-evolving job market.

8

Mission Innovate

Now, it's your turn! In Mission Innovate; students will apply their creativity, problem-solving skills, and knowledge to develop their own competition projects. This challenge encourages independent thinking, collaboration, and real-world innovation, empowering students to become the next generation of creators and problem-solvers.

Now I Can

This section is designed to help you measure your progress and achievements. Enjoy your exploration and learning journey through this STEAM book!

My Journal

In this part students will write their own story while they are creating the project.

By the end of this theme, students will be able to:

- Identify and empathize with the everyday challenges experienced by people with visual impairments.
- Describe the structure of the human eye and explain how it enables vision.
- Demonstrate an understanding of how the eye functions and what leads to vision loss.
- Use the Engineering Design Process to develop and refine solutions to real-world accessibility problems
- Explain how sound waves travel and how echolocation helps with spatial awareness.
- Investigate how ultrasonic sensors operate and how they are used to measure distance.
- Use estimation to quickly calculate and compare distances detected by sensors for design decisions.
- Assemble a basic electronic circuit that includes an Arduino, ultrasonic sensor, and buzzer.
- Write and modify Arduino code that triggers a buzzer response based on measured distance.
- Develop and test a mobile application that uses Bluetooth and text-to-speech to support users.
- Establish a Bluetooth connection between Arduino and the app to transmit sensor data.
- Design a 3D model of smart glasses using digital tools like Tinkercad.
- Enhance the prototype by adding additional sensors to improve functionality and safety.
- Use logical thinking to identify and fix errors related to measurement, scaling, and app.

Unit 1

Lesson 1

Spark The Challenge!

The Incident at The Fair.

Time Frame: [90 min].

Ready, Set, Learn!

By the end of this lesson students will be able to:

- Identify common challenges that individuals with visual impairments face in daily life
- Explain how people with visual impairments use other senses—like hearing, touch, and smell—to understand their surroundings and complete tasks.
- Describe examples of assistive technology and explain how these tools help people with visual impairments live more independently.
- Demonstrate empathy by describing how it might feel to navigate the world without sight and identifying the emotional and practical challenges involved.

Challenge questions of the lesson

1. **What would it be like to move through a crowded space without being able to see?**
2. **How can STEAM tools be used to help someone who can't rely on sight?**
3. **What senses become more important when one sense is missing?**
4. **How can we design technologies that respond to sound, touch, or smell instead of visuals?**
5. **Why is it important to understand the needs of people with disabilities when designing solutions?**

SEL Relation

- Empathy Building: Students imagine how it feels to move without sight, helping them understand the challenges faced by people with visual impairments.
- Perspective-Taking: Through activities like role-play and story reflection, students consider different points of view and lived experiences.
- Responsible Decision-Making: Students think critically about how to create helpful, inclusive solutions that make a positive impact.
- Collaboration & Communication: Group tasks, discussions, and coding projects promote teamwork, respectful listening, and idea sharing.
- Kindness & Inclusion: The lesson reinforces the value of designing for everyone and treating others with care and fairness.

Engage

Teacher Role:

- Spark curiosity and discussion using the four-part story as an emotional entry point.
- Scaffold understanding of disability by focusing on how Mr. Evans navigates the world without sight.
- Preview key vocabulary: blind, cane, traits, disability, safe, design.
- Encourage students to observe changes in Jamila's thoughts across the four scenes.
- Ask reflective questions after each scene and build a class-wide problem statement.

Expected From Students:

- Listen closely to the story and track how Jamila's feelings change.
- Reflect on Mr. Evans's experience using empathy.
- Share ideas about what makes walking safely difficult for someone who can't see.
- Help define a real-life challenge the class will try to solve.

Parts of the books included.


The school fair was full of energy and excitement. Stalls lined the pathways, and the air smelled like popcorn. Students were laughing, playing games, and looking at the exhibits.

Adam held up his remote-control car. "Look at my RC car! I can control it with my phone."

Laila stood next to him, showing off her robotic arm. "Check out my robotic arm. It's so cool!"

Jamila pointed toward the entrance. "Look, Mr. Evans is here with his dog!"

Mr. Evans walked over with a smile, his guide dog by his side. "Hello everyone, it's great to be here."

The fair was still busy when a loud noise—like a balloon popping—echoed through the fairground.

Mr. Evans jumped back as his guide dog got scared and ran into the crowd.

Mr. Evans: "Whoa! Where did he go?"

Adam looked around, worried. "His dog ran off!"

Laila grabbed Adam's arm. "We have to find him!"

STEAM SCOUTS 3


Pyramakerz®

The fair was still lively, but Adam, Laila, and Jamila were focused on finding the dog. Mr. Evans moved carefully, his hands out to feel his way.

Jamila turned to him. "Don't worry, Mr. Evans! We'll find your dog."

Laila pushed through the crowd. "It's hard to move with so many people here."

Adam glanced at Mr. Evans. "He's having trouble without his dog."

The group kept searching, determined to help."

A small crowd gathered as the principal stepped forward to speak. Mr. Evans stood nearby, looking thoughtful.

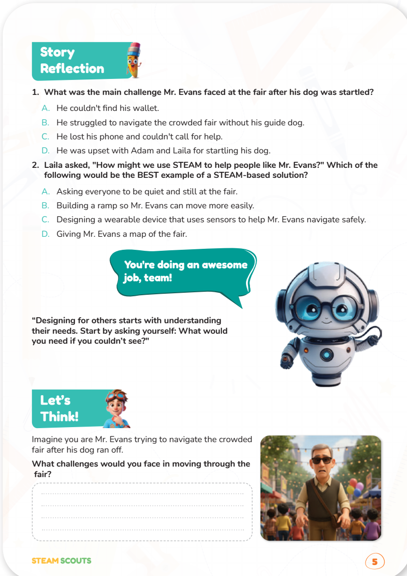
Principal: "Today, we saw how hard it can be for people with visual impairments to navigate busy places like this fair."

Laila raised her hand. "How can we use STEAM to help people like Mr. Evans?"

Adam nodded. "We need to learn more about how the eye works."

The students started sharing ideas, excited to use their skills to make a difference.

4 STEAM SCOUTS


Story Reflection

1. What was the main challenge Mr. Evans faced at the fair after his dog was startled?
   A. He couldn't find his wallet.
   B. He struggled to navigate the crowded fair without his guide dog.
   C. He lost his phone and couldn't call for help.
   D. He was upset with Adam and Laila for startling his dog.
2. Laila asked, "How might we use STEAM to help people like Mr. Evans?" Which of the following would be the BEST example of a STEAM-based solution?
   A. Asking everyone to be quiet and still at the fair.
   B. Building a ramp so Mr. Evans can move more easily.
   C. Designing a wearable device that uses sensors to help Mr. Evans navigate safely.
   D. Giving Mr. Evans a map of the fair.

You're doing an awesome job, team!

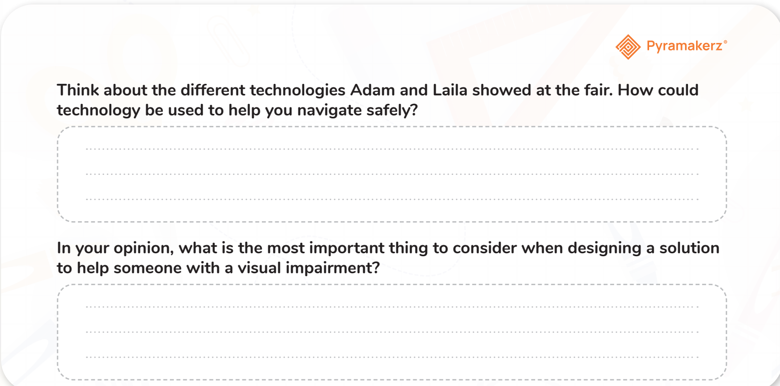
"Designing for others starts with understanding their needs. Start by asking yourself: What would you need if you couldn't see?"

Let's Think!

Imagine you are Mr. Evans trying to navigate the crowded fair after his dog ran off.

What challenges would you face in moving through the fair?

STEAM SCOUTS 5


Pyramakerz®

Think about the different technologies Adam and Laila showed at the fair. How could technology be used to help you navigate safely?

In your opinion, what is the most important thing to consider when designing a solution to help someone with a visual impairment?

Story

- Read the story scenes aloud or have students read in roles (Adam, Laila, Principal, etc.).
- Spark curiosity by pausing after Scene 2 and asking:
  - “How do you think Mr. Evans feels without his guide dog?”
- Guide students to empathize with Mr. Evans and observe how his experience inspired a design challenge.
- Scaffold the vocabulary (e.g., “visual impairment,” “navigate,” “STEAM”) to ensure understanding.
- Facilitate a quick pair-share after Scene 4:
  - “If you were in Laila’s shoes, what idea would you have to help Mr. Evans?”

Possible Strategies for Implementation:

1. **Scene-by-Scene Reflection.**
   - Strategy: Pause after each scene and ask short reflective questions.
     - Implementation:

       Scene 1: “What did Jamila notice about Mr. Evans?”

       Scene 2: “What clues showed Mr. Evans was blind?”

       Scene 3: “Why did Jamila want to help?”

       Scene 4: “What do you think Jamila might do next?”
2. **Empathy Roleplay (Light Version).**
   - Strategy: Reinforce understanding through movement and drama.
     - Implementation:
     1. Blindfold one student (with their consent) and ask them to walk slowly across a path with “safe” and “obstacle” markers.
     2. Others guide using only sounds (e.g., clap = stop, tap = turn).
     3. Ask: “What was difficult? What would have helped you feel safe?”
3. **See–Think–Wonder Routine.**
   - Strategy: Activate thinking about visibility and accessibility.
     - Implementation:
     1. Show an image of a busy sidewalk or market scene.
     2. Ask:
        - See: “What do you notice?”
        - Think: “What challenges might someone with a disability face here?”

- Wonder: “What might help them move around safely?”

4. **Whole-Class Problem Wall.**

- Strategy: Co-construct the unit’s big challenge.
  - Implementation:
  1. After the story, ask: “What’s the problem Mr. Evans faces?”
  2. Record student responses on sticky notes or the board.
  3. Finalize a version of the challenge together:

“How might we help Mr. Evans move around safely using science and technology?”

Story Reflection

Objective:

Ensure students comprehend the story and recognize how unexpected challenges—like losing a guide dog—can affect the emotions and safety of people with visual disabilities. Students will also connect this real-world problem to a meaningful STEAM-based solution.

Answer:

1. **What was the main challenge Mr. Evans faced at the fair after his dog was startled?**
   - Expected Answer: b) He struggled to navigate the crowded fair without his guide dog.
   - Reasoning: In the story, Mr. Evans’s guide dog runs off after being frightened. Without the dog, Mr. Evans has trouble moving through the noisy, busy fair. This shows how much he depends on the dog to stay safe and highlights the danger he faces when the dog is gone.
2. **Laila asked, "How might we use STEAM to help people like Mr. Evans?" Which of the following would be the BEST example of a STEAM-based solution?**
   - Expected Answer: c) Designing a wearable device that uses sensors to help Mr. Evans navigate safely.
   - Reasoning: STEAM (Science, Technology, Engineering, Arts, Math) focuses on solving real-world problems. Designing a sensor-based wearable device directly helps Mr. Evans move safely and is an example of using technology and innovation to support people with disabilities.

Possible Strategies for implementation:

1. **Guided Discussion with Visual Cues.**
   - Strategy: Help students identify Mr. Evans’s feelings through facial expressions and sound cues.
   - Implementation:
     - Display emotion cards (e.g., scared, confused, frustrated) or show a picture of Mr. Evans at the fair.
     - Ask: “Which face shows how Mr. Evans might have felt when his dog ran away?”

- Follow up: “Why do you think he felt that way?”

- Purpose: Supports emotional vocabulary and builds empathy through inference.

2. **Multiple Choice with Movement.**
   - Strategy: Engage students physically while reviewing comprehension.
   - Implementation:
     - Label four corners of the room A, B, C, and D.
     - Read the multiple-choice question aloud.
     - Students walk to the answer they think is correct.
     - Ask representatives from each corner to explain their reasoning, with teacher support as needed.
   - Purpose: Makes reflection interactive, inclusive, and multisensory.

Let’s Think!

Objective:

Students will reflect on the challenges faced by individuals with visual impairments in busy environments and brainstorm how technology can be used to design helpful solutions. This promotes empathy, problem-solving, and early design thinking.

Answers:

1. **What challenges would you face in moving through the fair?**
   - I might bump into people or objects because I can’t see them coming.
   - Loud noises could make it hard to hear where things are.
   - I would feel nervous not knowing what’s around me.
   - It might be confusing to know which way to go.
   - I could trip over things or get turned around in the crowd.
2. **How could technology be used to help you navigate safely?**
   - Sensors, like the ones on Adam and Laila’s projects, could detect obstacles.
   - The robotic arm could vibrate or make sounds to give directions.
   - A mobile app could tell me where it’s safe to walk.
   - Smart glasses could warn me if I get too close to someone or something.
   - Bluetooth technology could connect the glasses to a phone that gives voice instructions.
3. **What is the most important thing to consider when designing a solution to help someone with a visual impairment?**
   - It should be easy to use without needing to see.
   - It must give clear feedback, like sounds or vibrations.

- The device should be comfortable and not too heavy.
- It should help the person feel confident and safe.
- It must respect their independence and dignity.

Activity Instructions:

Step 1: Set the Stage.

- Say:
  - o “Imagine you are Mr. Evans at a busy fair, but your guide dog ran off! You can’t see clearly, and people are moving everywhere. How would you feel? What would be hard to do?”
- Optional: Show a crowded fairground image or play sound effects of a busy crowd.
- Ask:
  - o “What problems do you think Mr. Evans faced in that moment?”

Step 2: Activate Thinking.

- Review parts of the story showing how Mr. Evans struggled.
- Ask guiding questions:
  - o “What would be hard about moving around without a guide dog?”
  - o “What might help him feel safer and more confident?”

Step 3: Design Reflection Time (Student Work Time).

- Encourage:
  - o Detailed answers or drawings for each prompt.
  - o Imaginative solutions (e.g., sensors, beeps, wearable devices, or voice directions).
  - o Students to put themselves in Mr. Evans’s shoes when answering.
- Support students who prefer to draw their ideas using visual options like fair maps or device sketches.

Step 4: Share & Reflect.

- Invite students to present their ideas. Use reflection questions like:
  - o “What feature did you design that helps Mr. Evans know where to go?”
  - o “Why did you choose that technology?”
  - o “How would your design make someone feel safe or confident?”
- Display their work on a “Vision Wizards Design Wall.”

Explorer’s Toolkit

Teacher Role:

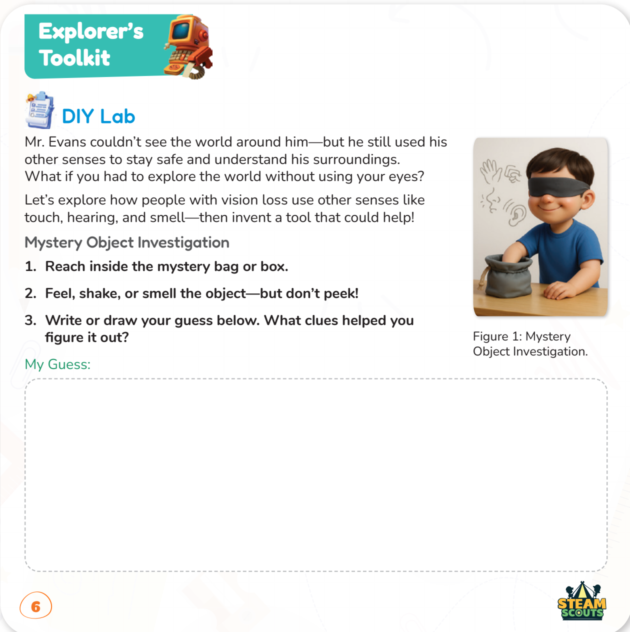
- **Facilitator:** Set up the “mystery bag” or boxes and guide students through each round of observation using non-visual clues.

- **Model:** Show how to gently feel or smell an object without peeking, and describe your own observations aloud.
- **Questioner:** Ask probing questions like "What helped you figure out what was inside?" or "What sense gave you the strongest clue?"
- **Support:** Assist students who may struggle with identifying textures, sounds, or scents, and encourage use of descriptive vocabulary.
- **Connector:** Reinforce how this activity connects to Mr. Evans's experience and real-life assistive tools like smart glasses.

Expected From Students:

- **Hands-On Participation:** Students will explore mystery objects using touch, sound, or smell only—no peeking!
- **Descriptive Thinking:** Students will describe what they felt, heard, or smelled using clear, sensory words.
- **Logical Guessing:** Students will record or draw their best guess about the object, supported by clues.
- **Empathy Connection:** Students will reflect on how it felt to explore without using sight and connect that to how Mr. Evans experiences the world.

Parts of the book included:


Explorer's Toolkit

DIY Lab

Mr. Evans couldn't see the world around him—but he still used his other senses to stay safe and understand his surroundings. What if you had to explore the world without using your eyes?

Let's explore how people with vision loss use other senses like touch, hearing, and smell—then invent a tool that could help!

Mystery Object Investigation

1. Reach inside the mystery bag or box.
2. Feel, shake, or smell the object—but don't peek!
3. Write or draw your guess below. What clues helped you figure it out?

Figure 1: Mystery Object Investigation.

My Guess:

6


What I felt or noticed:

DIY Lab

Objective:

Students will understand how people with visual impairments rely on alternative senses (touch, hearing, smell) to explore and interpret their environment. They will develop empathy and observation skills by identifying an object using only non-visual clues, building awareness of sensory-based problem-solving and inclusive design thinking.

Materials Needed:

- Mystery bags or boxes (1 per group or station).
- Everyday objects with varied textures, weights, and scents (e.g., sponge, bell, orange peel, paperclip, cotton ball, coin).
- Pencils or pens.

Activity Instructions:

Step 1: Introduction

- Say: “Mr. Evans couldn’t use his eyes to explore the world—but he used his other senses like touch and hearing. Today, we’ll try to do the same!”
- Emphasize: “No peeking—your job is to be a sense detective!”

Step 2: Mystery Sensing

- Each student or group reaches into a mystery bag.
- They feel, shake, or smell the object inside (without looking).
- They record their guess and what clues helped them.

Step 3: Class Reflection

- Ask:
  - “What sense helped you the most?”
  - “How confident were you without seeing?”
  - “How might this feel if you had to do it every day like Mr. Evans?”
- Encourage connections to real tools (e.g., white canes, smart sensors, talking signs).

Possible Strategies for Implementation:

1. **Set the Scene with Storytelling.**
   - How to Implement: Begin by reminding students of Mr. Evans’ experience at the fair from the story. Ask: “How did he know where to go? What senses do you think helped him most?” This builds context and curiosity.
2. **Mystery Bag Setup.**
   - How to Implement: Prepare several mystery bags or boxes with different objects (e.g., sponge, toy car, rubber ball, crumpled foil). Each object should be safe to touch, smell, or gently shake.
3. **Blindfold & Safety Protocol.**
   - How to Implement: Blindfold one student at a time or ask them to keep their eyes closed. Remind the class about respectful behavior and helping others. Monitor to ensure safety.
4. **Think Aloud Modeling.**
   - How to Implement: Model the activity first. Feel an object yourself and verbalize: “Hmm... it’s soft... maybe a sponge?” This encourages students to describe their reasoning clearly.

5. **Sensory Word Bank.**

- How to Implement: Provide or build a word bank with sensory words (e.g., rough, smooth, bumpy, squishy, sweet-smelling). This helps students articulate their observations.

6. **Reflection Share-Out.**

- How to Implement: After each guess, allow students to share:
  - What clues helped you figure it out?
  - Was it hard without your eyes?
  - How does this help us understand Mr. Evans' world?

7. **Empathy Connection**

- How to Implement: After the activity, lead a short discussion:
  - What kind of tools could we design to help people who can't see?
  - How can technology use these other senses to help?

Explain

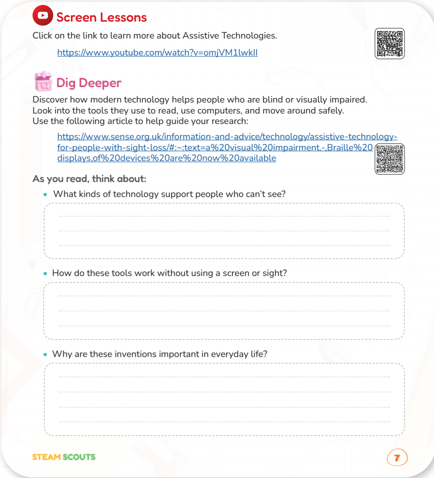
Teacher Role:

- Introduce key concepts such as accessibility, assistive technology, and how visual impairments affect daily life.
- Use guiding questions to activate prior knowledge and prepare students to engage with video and reading materials.
- Support comprehension by pausing for clarification, providing visual supports, and linking key points back to Mr. Evans's story.
- Lead group discussions that connect the screen lesson and article to real-life problems and student experiences.

Expected From Students:

- Watch the video attentively and identify examples of assistive technology.
- Read or listen to the article, focusing on how people who are blind use tech in daily life.
- Participate in group discussions and reflect on what they've learned.
- Share their thoughts using simple words, drawings, or technology sketches, and ask questions that connect learning to the challenge.

Parts of the books included.



Screen Lessons

Click on the link to learn more about Assistive Technologies.

https://www.youtube.com/watch?v=omjVM1lwkII

Dig Deeper

Discover how modern technology helps people who are blind or visually impaired.
Look into the tools they use to read, use computers, and move around safely.
Use the following article to help guide your research:

https://www.sense.org.uk/information-and-advice/technology/assistive-technology-for-people-with-sight-loss/#:~:text=a%20visual%20impairment.-,Braille%20displays,of%20devices%20are%20now%20available

As you read, think about:

- What kinds of technology support people who can't see?
- How do these tools work without using a screen or sight?
- Why are these inventions important in everyday life?

STEAM SCOUTS 7

Screen Lessons

Objective:

Students will explore the concept of assistive technologies and understand how these tools support individuals with disabilities in navigating daily life. By watching the video, they will identify real-world examples and gain insight into how technology can be designed with empathy and inclusivity in mind.

Video link:https://www.youtube.com/watch?v=omjVM1lwkII

Possible Strategies for implementation:

1. **Pre-Watching Strategy: Set a Purpose.**
   - Implementation:
     - Ask students:
       1. "What kinds of tools might help Mr. Evans see or move around safely?"
       2. "Have you ever used a tool to help you do something?"
     - Write responses on the board and say, "Let's see which of these ideas appear in the video!"
2. **During Watching Strategy: Notice and Note.**
   - Implementation:
     - Students jot down or sketch:
       1. One assistive technology shown.
       2. How it helps a person who is blind.
     - Encourage students to use symbols or quick sketches if writing is difficult.
3. **Post-Watching Strategy: Think-Pair-Share.**
   - Implementation:
     - Ask: "What surprised you? What new idea did you learn?"
     - Students think quietly → discuss with a partner → share with the class.

- Use prompts like:
  1. “How is this like or different from the smart glasses idea in our story?”
  2. “Would this tool help Mr. Evans at the fair?”

Dig Deeper

Objective:

Students will explore how people who are blind use modern technologies like computers, apps, and digital tools. They will learn how these tools work without sight and reflect on the importance of designing inclusive technology for everyday life.

Reading Resource: https://www.sense.org.uk/information-and-advice/technology/assistive-technology-for-people-with-sight-loss/#:~:text=a%20visual%20impairment.-,Braille%20displays,of%20devices%20are%20now%20available

Possible Strategies for implementation:

1. **Pre-Reading Strategy: Activate Curiosity.**
   - Implementation:
     - Show the article title and ask:
       1. “What do you already know about this?”
       2. “What do you wonder?”
     - Use a Know–Want to Know chart on the board. Fill in the “Learned” column after reading.
2. **During Reading Strategy: Stop & Sketch.**
   - Implementation:
     - Every 1–2 paragraphs, pause to let students:
       1. Sketch the tool mentioned (e.g., Braille display, screen reader).
       2. Write one key word or idea they learned.
     - Discuss as a class to review new terms like tactile feedback or voice control.
3. **Post-Reading Strategy: Summarize and Reflect.**
   - Implementation:
     - In pairs or small groups, students answer:
       1. “Which tool did you learn about?”
       2. “How does it work?”
       3. “How would it help someone like Mr. Evans?”
     - Students explain in their own words or drawings, and share their favorite invention.

Elaborate and Evaluate

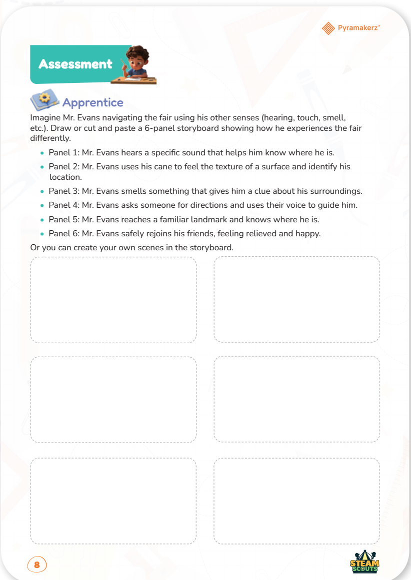
Teacher Role:

- Connect each activity to Mr. Evans' experience and the theme of using other senses.
- Model each task: demonstrate a storyboard panel, blindfold walk procedure, and sample MBlock code.
- Guide reflections with empathy-driven questions.
- Support students by offering prompts and troubleshooting help.
- Encourage creativity and inclusion in all designs and responses.

Expected From Students:

- **Apprentice:** Draw or create a 6-panel storyboard showing how Mr. Evans uses his senses at the fair.
- **Builder:** Participate in a blindfold walk, then reflect on the experience using the guiding questions.
- **Mastermind:** Code a Sound Alert Bot in MBlock, add a creative twist, and explain how it helps someone who can't see.

Parts of the book included:


Pyramakerz®

Assessment

Apprentice

Imagine Mr. Evans navigating the fair using his other senses (hearing, touch, smell, etc.). Draw or cut and paste a 6-panel storyboard showing how he experiences the fair differently.

- Panel 1: Mr. Evans hears a specific sound that helps him know where he is.
- Panel 2: Mr. Evans uses his cane to feel the texture of a surface and identify his location.
- Panel 3: Mr. Evans smells something that gives him a clue about his surroundings.
- Panel 4: Mr. Evans asks someone for directions and uses their voice to guide him.
- Panel 5: Mr. Evans reaches a familiar landmark and knows where he is.
- Panel 6: Mr. Evans safely rejoins his friends, feeling relieved and happy.

Or you can create your own scenes in the storyboard.

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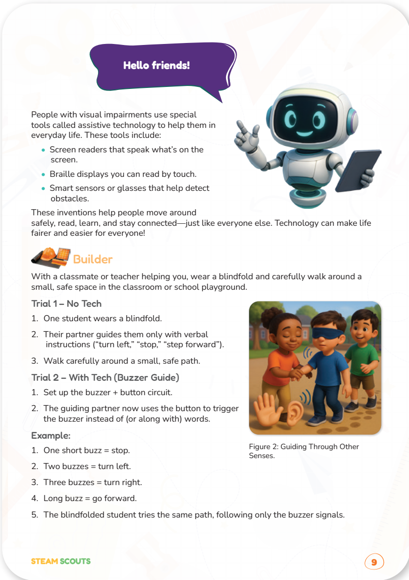

Hello friends!

People with visual impairments use special tools called assistive technology to help them in everyday life. These tools include:

- Screen readers that speak what's on the screen.
- Braille displays you can read by touch.
- Smart sensors or glasses that help detect obstacles.

These inventions help people move around safely, read, learn, and stay connected—just like everyone else. Technology can make life fairer and easier for everyone!

Builder

With a classmate or teacher helping you, wear a blindfold and carefully walk around a small, safe space in the classroom or school playground.

Trial 1 – No Tech

1. One student wears a blindfold.
2. Their partner guides them only with verbal instructions ("turn left," "stop," "step forward").
3. Walk carefully around a small, safe path.

Trial 2 – With Tech (Buzzer Guide)

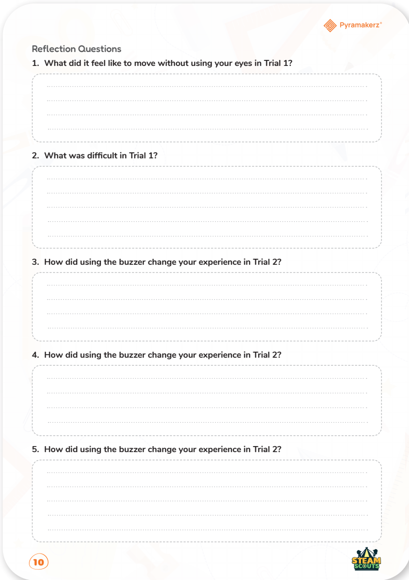
1. Set up the buzzer + button circuit.
2. The guiding partner now uses the button to trigger the buzzer instead of (or along with) words.

Example:

1. One short buzz = stop.
2. Two buzzes = turn left.
3. Three buzzes = turn right.
4. Long buzz = go forward.
5. The blindfolded student tries the same path, following only the buzzer signals.

Figure 2: Guiding Through Other Senses.

STEAM SCOUTS 9


Pyramakerz®

Reflection Questions

1. What did it feel like to move without using your eyes in Trial 1?
2. What was difficult in Trial 1?
3. How did using the buzzer change your experience in Trial 2?
4. How did using the buzzer change your experience in Trial 2?
5. How did using the buzzer change your experience in Trial 2?

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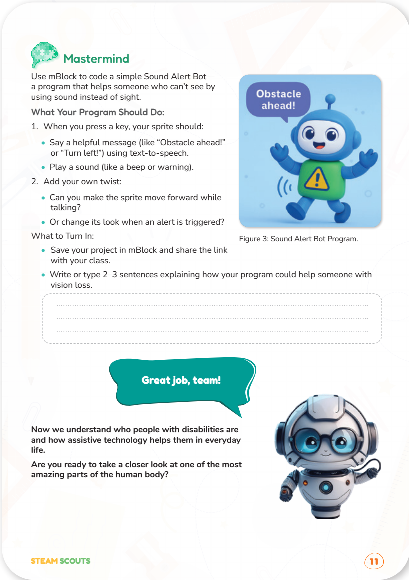

Mastermind

Use mBlock to code a simple Sound Alert Bot—a program that helps someone who can't see by using sound instead of sight.

What Your Program Should Do:

1. When you press a key, your sprite should:
   - Say a helpful message (like "Obstacle ahead!" or "Turn left!") using text-to-speech.
   - Play a sound (like a beep or warning).
2. Add your own twist:
   - Can you make the sprite move forward while talking?
   - Or change its look when an alert is triggered?

Figure 3: Sound Alert Bot Program.

What to Turn In:

- Save your project in mBlock and share the link with your class.
- Write or type 2–3 sentences explaining how your program could help someone with vision loss.

Now we understand who people with disabilities are and how assistive technology helps them in everyday life.

Are you ready to take a closer look at one of the most amazing parts of the human body?

STEAM SCOUTS 11

Apprentice

Objective:

Students will use empathy and creative thinking to imagine how Mr. Evans explores the fair using non-visual senses (hearing, touch, smell, etc.). They will illustrate his experience in a 6-panel storyboard, demonstrating understanding of sensory awareness, accessibility, and inclusive design thinking.

Materials Needed:

- Blank storyboard templates (provided in book or teacher printouts) or plain paper.
- Crayons, colored pencils, or markers.
- (Optional) Scissors and magazines for cut-and-paste visuals.
- Sentence starters:
  - o "Mr. Evans hears..."
  - o "The ground feels..."
  - o "He smells..."
  - o "He asks someone..."
  - o "He remembers..."

You can use this as a storyboard template :

https://drive.google.com/file/d/1m0MRQHOBH_10slAsFLsKzA4-s1-fbp_E/view?usp=sharing

Activity Instructions:

1. Understand the Task: A Day at the Fair – Mr. Evans' Senses.

- Begin by reviewing the story scenes about Mr. Evans navigating the crowded school fair.
- Explain that students will now imagine the fair through Mr. Evans' perspective—someone who is visually impaired.

- Discuss how people with vision loss use their other senses (hearing, touch, smell, etc.) to move safely and understand their surroundings.

2. Explore the Prompt: What Might Mr. Evans Experience?

- Present the 6-panel storyboard worksheet.
- Walk students through each panel description, asking them what Mr. Evans might hear, smell, feel, or ask for.
- Encourage students to also brainstorm their own custom versions of the story if they want to replace any of the panels.

3. Draw or Build the Storyboard: Show the Journey.

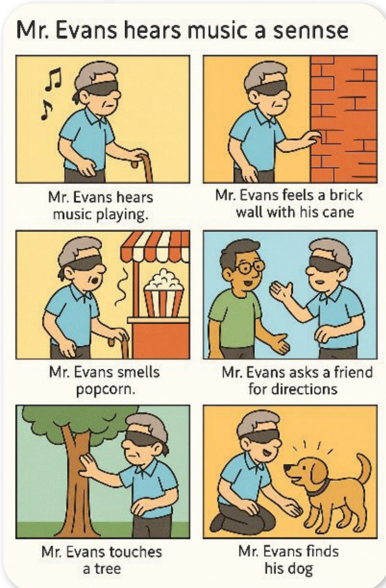
- Students may either draw or cut-and-paste images (if provided) to complete each panel.
- Remind them to label which sense is being used in each scene (e.g., "he hears music," "he feels the texture of the path").
- Guide them to use arrows or short captions to show how Mr. Evans navigates from one point to another.

4. Observe and Reflect: Mr. Evans' Experience.

- After students complete the storyboard, prompt them with questions to deepen their understanding:
  - o What sense was most helpful to Mr. Evans in your story?
  - o What challenges did he face along the way?
  - o If you were helping him at the fair, what would you do?

5. Present and Share: Stories of the Senses.

- Invite students to present their storyboard to a partner or the class.
- Encourage them to explain their drawings or scene choices: What sense is being used? How did it help Mr. Evans?
- Celebrate creative additions or student-invented assistive tools drawn in the storyboards.



Possible Strategies for Implementation:

1. Visual Brainstorm Before Drawing.

- Display or describe fairground scenes (e.g., popcorn stand, music, gravel paths).
- Ask students:
  - o "If you closed your eyes, what would you notice first?"
  - o "Which sense would help you find your way?"
- Chart student answers on the board to reference during drawing.

2. Guided Storyboard Creation.

- Offer a sample sequence of six scenes or let students customize their own journey.

- Encourage labeling under each panel for students who need support with drawing.
- Let students work solo or collaborate in pairs (one draws, one describes).

3. **Reflection and Sharing Circle.**

- Once complete, hold a “gallery walk” or partner swap to view each other’s work.
- Discussion questions:
    - “Which sense helped Mr. Evans the most in your story?”
    - “What would be hardest if you couldn’t see?”
    - “What kind of smart invention could help in this situation?”

Builder

Objective:

- Develop student empathy for people with visual impairments.
- Explore how simple circuits (buzzer + button) can serve as assistive tools.
- Compare experiences with and without technology support.
- Encourage reflection on how AI could enhance accessibility tools.

Materials Needed:

- Blindfolds (1 per pair/group).
- Buzzer + battery + push button circuit (or breadboard with wires/components).
- Safe walking space (classroom path or playground area with soft obstacles).
- Reflection worksheet (provided in student book).

Activity Instructions:

1. Set the Stage.

- Explain to students that they will experience what it’s like to move without sight and then test how a simple buzzer circuit can act as an assistive tool.
- Remind students that empathy and safety are the most important parts of this activity.

2. Trial 1 – No Tech.

- Divide students into pairs (guide + blindfolded student).
- Have the guide give only verbal directions (e.g., “turn left,” “stop,” “step forward”).
- Allow the blindfolded student to carefully walk through the safe space.

3. Trial 2 – With Tech (Buzzer Guide).

- Demonstrate how the buzzer + button circuit works.
- Agree on simple buzzer “codes” (e.g., 1 buzz = stop, 2 buzzes = left, 3 buzzes = right, long buzz = forward).
- Repeat the walk with the blindfolded student, but now the guide uses only the buzzer signals instead of words.

4. Reflection

- After both trials, ask students to complete the reflection questions.
- Facilitate a class discussion: Which method was easier? What challenges came up? How might AI improve this system?

5. Wrap Up

- Reinforce that assistive technologies help people with visual impairments live more independently.
- Highlight that AI can make these tools “smarter” by recognizing obstacles or giving spoken instructions automatically.

Possible Answers to Reflection Questions.

1. What did it feel like to move without using your eyes in Trial 1?

- Nervous, slow, unsure, unsafe.

2. What was difficult in Trial 1?

- Couldn’t see obstacles, hard to trust directions, fear of bumping into things.

3. How did using the buzzer change your experience in Trial 2?

- Easier to follow, quicker signals, sometimes confusing to remember buzz codes.

4. Which method (voice vs. buzzer) felt easier or safer? Why?

- Voice is easier to understand.
- Buzzer feels faster and can be discreet.
- Both have advantages.

5. How could this simple buzzer system be improved with AI in the future?

- Add sensors so it detects obstacles automatically.
- Add voice instructions instead of buzzes.
- Personalize signals to match walking speed.

Possible Strategies for Implementation.

- **Pair/Group Work:** Students work in pairs—one blindfolded, one guiding. Switch roles so each experiences both.
- **Safety Considerations:** Choose a controlled, obstacle-free space. Use soft objects (cones, chairs, mats) to simulate barriers.
- **Discussion:** After the activity, hold a class conversation on feelings, challenges, and tech’s role in accessibility.
- **Extension Activity:** Ask students to brainstorm improvements using AI (object detection, voice alerts, smart glasses).
- **Real-World Connection:** Show students an example of AI-powered assistive tech (short video demo of smart glasses or screen readers).

Mastermind

Objective:

- Students will apply their understanding of assistive technology and programming to design a simple sound-based alert system using mBlock that supports individuals with visual impairments in navigating safely.

Activity Instructions:

1. Introduce the Task (During Class):

- Tell students that their next mission is to build a "Sound Alert Bot" in mBlock to help people who can't see.
- Explain that this bot should use sound and speech instead of sight to alert the user to obstacles.
- Review the two essential program features:
    - o Use text-to-speech to say messages like "Obstacle ahead!" or "Turn left!"
    - o Add a sound effect like a beep or buzzer

2. Add Creativity (Encourage Extension):

- Ask students: "Can you make the bot move forward while it talks?"
- Or: "What if the sprite changed its look when it gave an alert?"

3. Assign as Homework:

- Instruct students to complete this coding task at home.
- Tell them to save their project in mBlock and bring a working link or screenshot to class.
- Students must also write 2–3 short sentences explaining how their program could support someone with vision loss.

Possible Strategies for Implementation.

- Strategy: "Demo & Design Challenge"
    - o How to Implement:

1. Demo Time (during class): Quickly show a sample sprite on mBlock saying "Watch out!" when a key is pressed.
2. Ask: "What would this feel like if you couldn't see the screen?"
3. Challenge students to design their own version with unique alert messages, sounds, or movement.
4. Send students home with a printed instruction sheet or video link for step-by-step support.

- Strategy: "User First" – Empathy Mapping Before Coding.
    - o How to Implement:

1. Ask students: “Imagine being in a crowded place like Mr. Evans. What kind of alert would help you know where to go?”
2. Create a quick “Help Me Navigate” list on the board (e.g., stop signs, left/right cues, sounds).
3. Have students choose one helpful idea and build it into their sprite’s voice or sound behavior in mBlock.

Now I Can

Objective:

Students will reflect on their learning by identifying how people with visual impairments navigate the world. Through the checklist, students will demonstrate empathy, understanding of assistive technologies, and awareness of how other senses support movement and independence.

Activity Instructions:

Step 1: Revisit the Experience.

- Say:
    - o “You’ve explored how Mr. Evans moved through the fair without using his eyes. Now let’s think about all the things you’ve learned!”
- Display or distribute the ‘Now I Can’ checklist and read it aloud with the class.

Step 2: Student Self-Reflection.

- Ask students to open their checklist and use a pencil or crayon.
- Instruct them to:
    - o Check what they feel confident about.
    - o Draw or write one thing they learned.
    - o Say: “Pick one thing and show what you learned—by drawing it or writing about it.”

Step 3: Small Group Discussion.

- In groups of 3–4, students take turns sharing:
    - o “What did you check off?”
    - o “What did you do that helped you learn that?”
    - o “Why is that important for helping people like Mr. Evans?”

Relation to Standards.

1. NGSS (Next Generation Science Standards).

- **3-PS2-1**

Standard: Plan and conduct an investigation to provide evidence of the effects of balanced and unbalanced forces on the motion of an object.

Relation: Students explore how individuals who are blind detect movement using sound or touch, simulating sensory input that helps guide motion.

- **3-PS2-4**

Standard: Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints on materials, time, or cost.

In this lesson:

  - Students consider the needs of people with vision impairment.
  - They brainstorm and propose assistive tech solutions to support mobility and navigation.

- **3-5-ETS1-1 (Engineering Design).**

Standard: Define a simple design problem reflecting a need or a want that includes specified criteria for success and constraints.

In this lesson:

  - Students identify a real-world problem (navigating a crowded fair with vision loss).
  - They define design needs for technologies like smart glasses or sound bots.

- **3-5-ETS1-2**

Standard: Generate and compare multiple possible solutions to a problem based on how well each is likely to meet the criteria and constraints of the problem.

In this lesson:

  - Students explore multiple assistive solutions (storyboard, sound alert bot, cane, voices).
  - They reflect on the strengths and challenges of each sensory strategy.

2. ISTE Standards for Students.

- **Empowered Learner (1a, 1c).**
  - Students set personal learning goals and reflect on their understanding using digital tools.

In this lesson: Students explore assistive tech in real life and simulate use through role-play and coding challenges.

- **Innovative Designer (4a, 4b, 4c).**
  - Students identify real-world problems, use the design process to imagine solutions, and test them through digital prototypes.

In this lesson: Students design a storyboard to visualize needs, and code a Sound Alert Bot using mBlock.

- **Creative Communicator (6a, 6b).**
  - Students choose digital tools to share ideas and present information clearly.

In this lesson: Students use visual storytelling and programming to explain how assistive tools help with vision loss.